

2.

(a)

(1) Instrument Relevance

(2) Instrument Exogeneity

(b) Note: A detailed explanation can be found in the lecture and tutorial videos.

Since $Cov(z, u) = 0$, the covariance relationship between z and y is

$$Cov(z, y) = Cov(z, \beta_0 + \beta_1 x + u) = \beta_1 Cov(z, x) + Cov(z, u) = \beta_1 Cov(z, x).$$

If we solve for β_1 , $\beta_1 = \frac{Cov(z, y)}{Cov(z, x)}$

$$\hat{\beta}_{1,IV} = \frac{\widehat{Cov(z, y)}}{\widehat{Cov(z, x)}} = \frac{\frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z})(y_i - \bar{y})}{\frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z})(x_i - \bar{x})} = \frac{\sum_{i=1}^n (z_i - \bar{z})y_i}{\sum_{i=1}^n (z_i - \bar{z})x_i} \text{ and } \hat{\beta}_{0,IV} = \bar{y} - \hat{\beta}_{1,IV} \bar{x}$$

$$\begin{aligned} \hat{\beta}_{1,IV} &= \frac{\sum_{i=1}^n (z_i - \bar{z})y_i}{\sum_{i=1}^n (z_i - \bar{z})x_i} \\ &= \frac{\sum_{i=1}^n (z_i - \bar{z})(\beta_0 + \beta_1 x_i + u_i)}{\sum_{i=1}^n (z_i - \bar{z})x_i} \\ &= \frac{\beta_0 \sum_{i=1}^n (z_i - \bar{z})}{\sum_{i=1}^n (z_i - \bar{z})x_i} + \frac{\beta_1 \sum_{i=1}^n (z_i - \bar{z})x_i}{\sum_{i=1}^n (z_i - \bar{z})x_i} + \frac{\sum_{i=1}^n (z_i - \bar{z})u_i}{\sum_{i=1}^n (z_i - \bar{z})x_i} \\ &= \beta_1 + \frac{\sum_{i=1}^n (z_i - \bar{z})u_i}{\sum_{i=1}^n (z_i - \bar{z})x_i} \end{aligned}$$

(c) Note: A detailed explanation can be found in the lecture and tutorial videos.

$$\begin{aligned}
 \hat{\beta}_{1,IV} &= \frac{\sum_{i=1}^n (z_i - \bar{z}) y_i}{\sum_{i=1}^n (z_i - \bar{z}) x_i} \\
 &= \frac{\sum_{i=1}^n (z_i - \bar{z}) (\beta_0 + \beta_1 x_i + u_i)}{\sum_{i=1}^n (z_i - \bar{z}) x_i} \\
 &= \frac{\beta_0 \sum_{i=1}^n (z_i - \bar{z})}{\sum_{i=1}^n (z_i - \bar{z}) x_i} + \frac{\beta_1 \sum_{i=1}^n (z_i - \bar{z}) x_i}{\sum_{i=1}^n (z_i - \bar{z}) x_i} + \frac{\sum_{i=1}^n (z_i - \bar{z}) u_i}{\sum_{i=1}^n (z_i - \bar{z}) x_i} \\
 &= \beta_1 + \frac{\sum_{i=1}^n (z_i - \bar{z}) u_i}{\sum_{i=1}^n (z_i - \bar{z}) x_i} \\
 &= \beta_1 + \frac{\frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z}) u_i}{\frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z}) x_i} \approx \beta_1
 \end{aligned}$$

When n goes to infinity, it is the population covariance $Cov(z, u) = 0$

(d) Note: A detailed explanation can be found in the lecture and tutorial videos.

$$\hat{\beta}_{1,IV} = \frac{\sum_{i=1}^n (z_i - \bar{z}) y_i}{\sum_{i=1}^n (z_i - \bar{z}) x_i} = \beta_1 + \frac{\sum_{i=1}^n (z_i - \bar{z}) u_i}{\sum_{i=1}^n (z_i - \bar{z}) x_i}$$

$$\begin{aligned} Var(\hat{\beta}_{1,IV}) &\approx \frac{\sigma^2 \sum_{i=1}^n (z_i - \bar{z})^2}{(\sum_{i=1}^n (z_i - \bar{z}) x_i)^2} \\ &= \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \times \frac{(\sum_{i=1}^n (z_i - \bar{z})^2)(\sum_{i=1}^n (x_i - \bar{x})^2)}{(\sum_{i=1}^n (z_i - \bar{z}) x_i)^2} \\ &= Var(\hat{\beta}_{1,OLS}) \times \frac{1}{\hat{\rho}_{z,x}^2} \geq Var(\hat{\beta}_{1,OLS}) \end{aligned}$$

$$\text{where } \hat{\rho}_{z,x}^2 = \frac{\left(\frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z}) x_i\right)^2}{\left(\frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z})^2\right) \left(\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2\right)} \leq 1$$

3(a)

. reg lwage sibs

Source	SS	df	MS	Number of obs	=	935
Model	3.86818074	1	3.86818074	F(1, 933)	=	22.31
Residual	161.788103	933	.173406326	Prob > F	=	0.0000
				R-squared	=	0.0234
				Adj R-squared	=	0.0223
Total	165.656283	934	.177362188	Root MSE	=	.41642

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
sibs	<u>-.0279044</u>	.0059082	-4.72	0.000	-.0394992	-.0163096
_cons	6.861076	.0220776	310.77	0.000	6.817748	6.904403

There is a negative relationship between log(wage) and sibs. The circled number is not return to education. Please refer to the video for more explanation.

(b)

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. reg educ brthord
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Source	SS	df	MS	Number of obs	=	852
Model	173.087012	1	173.087012	F(1, 850)	=	37.29
Residual	3945.88364	850	4.64221605	Prob > F	=	0.0000
				R-squared	=	0.0420
				Adj R-squared	=	0.0409
Total	4118.97066	851	4.84015353	Root MSE	=	2.1546

educ	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
brthord	<u>-.2826441</u>	.0462882	-6.11	0.000	-.3734967	-.1917915
_cons	14.14945	.1286754	109.96	0.000	13.89689	14.40201

It *could be* that older children are given priority for higher education, and families may hit budget constraints and may not be able to afford as much education for children born later.

(c) π_1 cannot be zero.

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. reg educ brthord sibs
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Source	SS	df	MS	Number of obs	=	852
Model	240.246365	2	120.123183	F(2, 849)	=	26.29
Residual	3878.72429	849	4.56857985	Prob > F	=	0.0000
				R-squared	=	0.0583
				Adj R-squared	=	0.0561
Total	4118.97066	851	4.84015353	Root MSE	=	2.1374

educ	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
brthord	-.1526742	.0570764	-2.67	0.008	-.2647017	-.0406467
sibs	-.1528673	.0398705	-3.83	0.000	-.2311236	-.0746109
_cons	14.2965	.1332881	107.26	0.000	14.03489	14.55811

-2.67 is smaller than the critical value -1.96 → Reject the null

P-value 0.008 is smaller than a 5% significance level → Reject the null

(d)

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. ivreg lwage (educ= brthord) sibs
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Instrumental variables (2SLS) regression

Source	SS	df	MS	Number of obs	=	852
Model	-7.96903706	2	-3.98451853	F(2, 849)	=	10.90
Residual	154.785102	849	.182314607	Prob > F	=	0.0000
Total	146.816065	851	.172521815	R-squared	=	.
				Adj R-squared	=	.
				Root MSE	=	.42698

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.1369941	.0746812	1.83	0.067	-.0095873	.2835756
sibs	.0021108	.0173717	0.12	0.903	-.0319858	.0362073
_cons	4.938527	1.05569	4.68	0.000	2.866458	7.010596

Instrumented: educ

Instruments: sibs brthord

Positive relationship between educ and lwage; no significant relationship between sibs and lwage.

The standard errors are bigger compared to the case that does not use IV.

WRONG (OLS) Model

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. reg lwage educ sibs
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Source	SS	df	MS	Number of obs	=	935
Model	17.210155	2	8.60507751	F(2, 932)	=	54.03
Residual	148.446128	932	.159276962	Prob > F	=	0.0000
				R-squared	=	0.1039
				Adj R-squared	=	0.1020
Total	165.656283	934	.177362188	Root MSE	=	.3991

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0560375	.0061227	9.15	0.000	.0440216	.0680534
sibs	-.0151325	.0058318	-2.59	0.010	-.0265774	-.0036876
_cons	6.068773	.0891164	68.10	0.000	5.893881	6.243665

(e)

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. reg educ brthord sibs
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Source	SS	df	MS	Number of obs	=	852
Model	240.246365	2	120.123183	F(2, 849)	=	26.29
Residual	3878.72429	849	4.56857985	Prob > F	=	0.0000
				R-squared	=	0.0583
				Adj R-squared	=	0.0561
Total	4118.97066	851	4.84015353	Root MSE	=	2.1374

educ	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
brthord	-.1526742	.0570764	-2.67	0.008	-.2647017	-.0406467
sibs	-.1528673	.0398705	-3.83	0.000	-.2311236	-.0746109
_cons	14.2965	.1332881	107.26	0.000	14.03489	14.55811

```
. predict educ_hat  
(option xb assumed; fitted values)  
(83 missing values generated)
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```
. corr educ_hat sibs  
(obs=852)
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	educ_hat	sibs
educ_hat	1.0000	
sibs	-0.9295	1.0000

Refer to the video for more information.